

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every ~7 Months

Headline finding: The 50% task-completion time horizon for frontier AI agents has doubled roughly every 7 months since 2019, rising from ~2 seconds (GPT-2) to ~1 hour 50 minutes (o3).

A New Metric: 50% Task-Completion Time Horizon

1 Assemble diverse task suite

Curate 170 software / research tasks spanning 1 second to 8 hours of skilled human effort.

2 Measure humans & AI agents

Time skilled human professionals on each task; run AI agents and record success rates.

3 Fit logistic model

Estimate the task duration at which the AI achieves 50% success — the time horizon.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Figure: Methodology overview — task suite construction, human baseline timing, and logistic model fitting.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Suite	Tasks	Human Time Range	Description
HCAST	97	1 min – 30 hrs	Diverse software engineering tasks (debugging, refactoring, data munging).
RE-Bench	7	~8 hrs each	Difficult ML research engineering tasks (e.g., custom CUDA kernels).
SWAA	66	1 sec – 30 sec	Short single-step atomic actions (e.g., identify shell scripts).

Example Tasks by Human Duration

Task Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: Which file is a shell script?
wikipedia_research	1 minute	Research simple factual information from Wikipedia.
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files.
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules.
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30× speedup of a backtesting tool.

Over 800 human baselines collected (2,529 total hours).

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes

(x2) in 4 Years

~207 days — Doubling time

$R^2 = 0.97$ — Fit quality

~7 months — Period

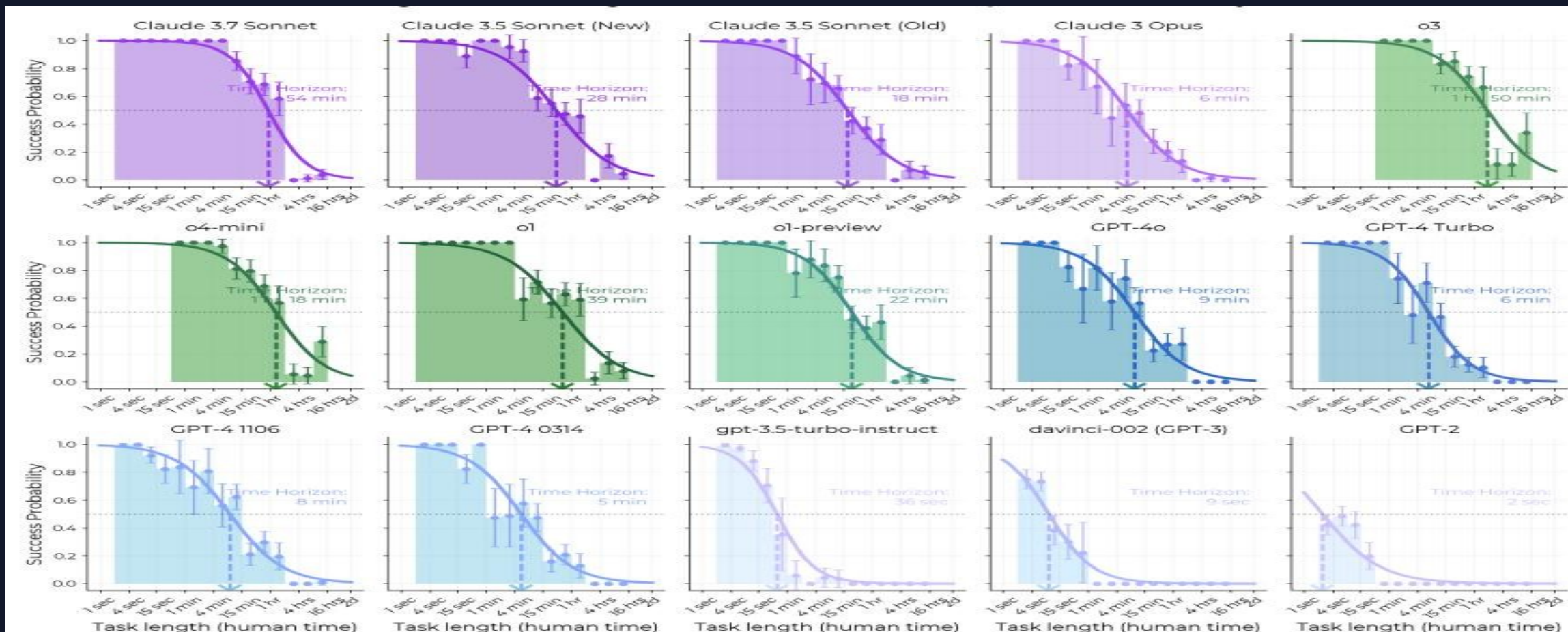


Figure: Exponential trend in 50% time horizon vs. model release date. Key milestones: GPT-2 (~2 sec), GPT-4 (~5 min), o3 (~1 hr 50 min).

Success Probability Drops Sharply as Task Length Increases for All Models

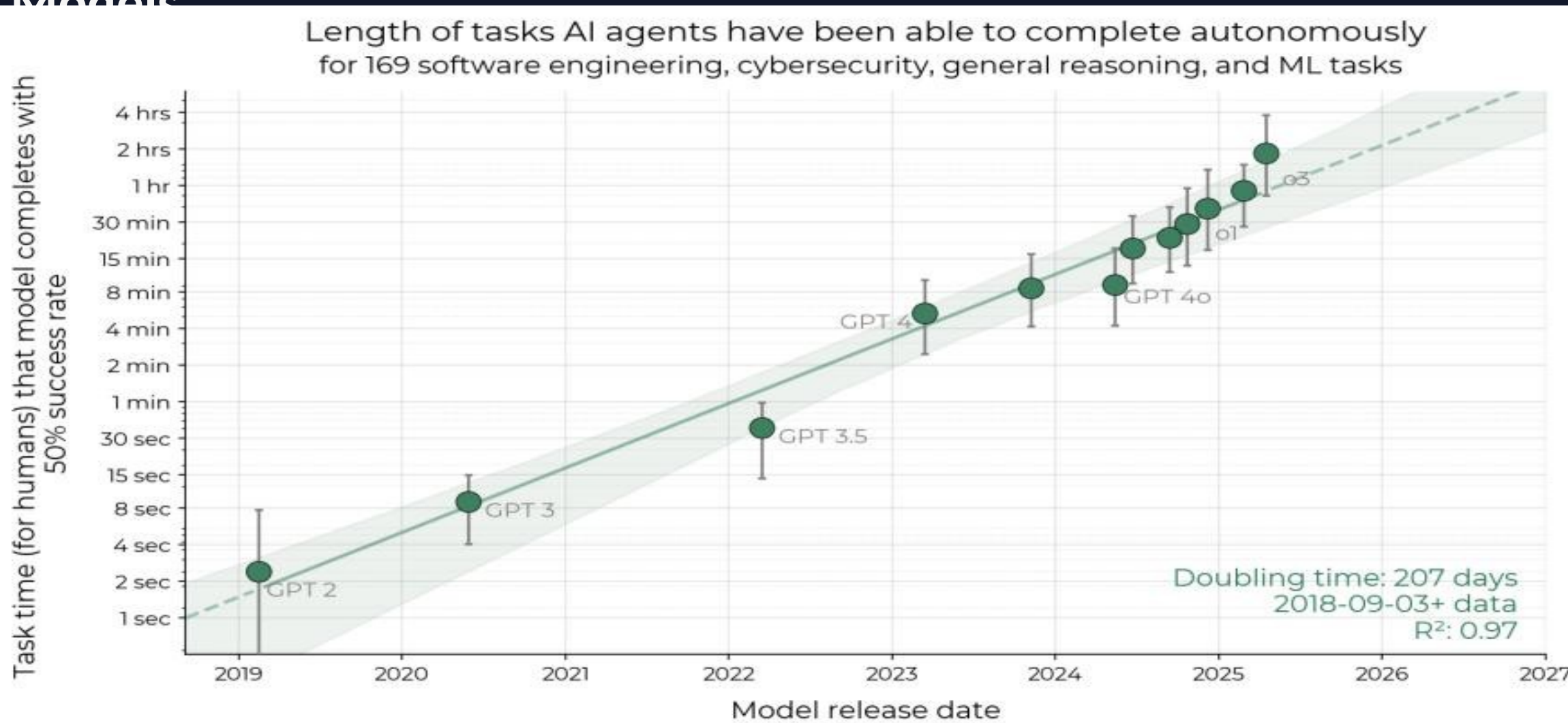


Figure: Per-model success probability curves. Frontier models (Claude 3.7 Sonnet, o3) push the curve rightward; older models (GPT-2, GPT-3) drop to near-zero within seconds.

Reasoning, Tool Use, and Error Recovery Drive Time Horizon

Growth

Key Capability Drivers

- Improved logical reasoning — longer chains of inference without degradation.
- Better tool use — effective integration of IDEs, terminals, libraries, and APIs.
- Greater reliability & self-awareness — knowing when to stop, verify, or ask.
- Ability to adapt to mistakes — detecting errors and backtracking autonomously.

Limitations & Open Questions

- Performance is notably lower on less structured, “messier” tasks.
- External validity remains uncertain — tasks may not perfectly represent real-world intellectual labor.
- Supplementary experiments found little evidence that trends are slower on more realistic tasks, but further validation is needed.

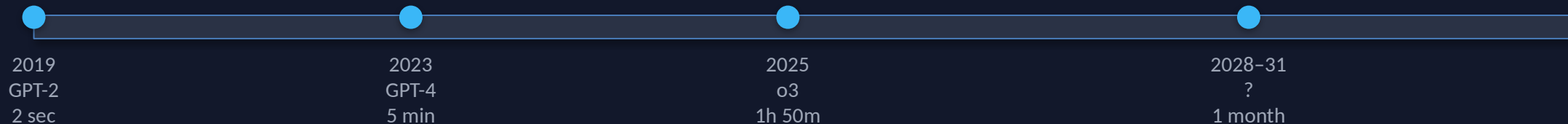
Insight: The gap between structured benchmark tasks and messy real-world work is the critical unknown. Closing it could accelerate—or complicate—forecasts.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

1 month time horizon

≈ 167 work hours | mid-2028 to mid-2031

- Naive exponential extrapolation of the 2019–2025 trend reaches >1 month horizon between mid-2028 and mid-2031.
- The 2023–2025 growth rate is ~20% faster than the overall 6-year rate, suggesting possible acceleration.
- 80% time horizon follows a similar trend but is roughly 5× shorter than the 50% horizon.
- Caveats: external validity uncertain; trend may bend due to data scarcity, compute limits, or algorithmic plateaus.



Takeaways: Exponential Autonomous Capability Growth Demands

Proactive Safety Governance

1. Robust metric — The 50% time horizon is an intuitive, human-grounded measure of AI autonomy with $R^2 = 0.97$ over 6 years of releases.
2. Rapid doubling — The horizon doubles every ~7 months; the 2023–2025 rate is ~20% faster, hinting at acceleration.
3. Current frontier — o3 autonomously completes tasks requiring nearly 2 hours of skilled human work, up from 2 seconds for GPT-2.
4. External validity open — Benchmark tasks may not perfectly represent real-world intellectual labor; generalization remains uncertain.
5. Governance imperative — If trends hold, month-long software project automation arrives by 2028–2031. Safety frameworks, monitoring, and policy must outpace capability growth.